

Joanie Hayoun Chung

jchung02@korea.ac.kr | github.com/jchung02

EDUCATION

M.S. in Statistics, Korea University

Mar. 2026 – Aug. 2027(Expected)

B.S. in Business Administration & Statistics, Korea University

Mar. 2021 – Feb. 2026

GPA: 4.22/4.5

Seoul, Korea

Copenhagen Business School

Aug. 2023 – Dec. 2023

Exchange Student

Copenhagen, Denmark

PUBLICATIONS

ORCA: Orchestrating Causal Agent

- Joanie Hayoun Chung, Chaemyung Lim, Sumin Lee, Sungbin Lim; *CHI 2026 Posters*.

Score-informed Neural Operator for Enhancing Ordering-based Causal Discovery

- Jiyeon Kang, Songseong Kim, Chanhui Lee, Doyeong Hwang, Joanie Hayoun Chung, Yunkyoung Ko, Sumin Lee, Sungwoong Kim, Sungbin Lim; *NeurIPS 2025*.

WORK EXPERIENCE

AGI Lab, under Prof. Sungbin Lim

Jan. 2025 – Feb. 2026

Student Researcher / Intern

Seoul, Korea

- Led the design and implementation of ORCA, an interactive multi-agent framework for causal analysis on relational databases; published at CHI 2026 Posters.
- Contributed to research on score-informed neural operators for ordering-based causal discovery; published at NeurIPS 2025.
- Presented in lab seminars on probabilistic circuits, GFlowNets, LLM post-training, and distributed training.

Korea University Business School, under Prof. Woochan Kim

Sep. 2024 – Feb. 2025

Research Assistant

Seoul, Korea

- Developed an LLM-based system to extract corporate governance indicators—such as shareholder rights and board structure—from unstructured disclosures across 800 KOSPI/KOSDAQ-listed firms (2000–2023), using data from DART and KIND.
- Automated large-scale crawling and parsing of annual and shareholder reports, and standardized variable definitions across evolving document formats to ensure 24-year longitudinal consistency.

PROJECTS

ORCA: Multi-Agent Framework for Causal Analysis on Relational Databases | LangGraph · PostgreSQL · LLM APIs

- Designed ORCA, a multi-agent framework that restructures LLM-assisted causal analysis with human-in-the-loop checkpoints, addressing a bottleneck where analysts spend 77% of time on repetitive data exploration and baseline LLM analyses produce reliable estimates only 8% of the time.
- Integrated specialized agents for data exploration, causal discovery, and causal inference using LangGraph with shared state; implemented diagnostics-driven algorithm selection, bootstrap stability scoring, and DoWhy-based treatment effect estimation across an 18-table, 130-variable relational database.
- Evaluated ORCA via a controlled user study (N=12) showing 76% reduction in exploration time, 42%p higher task completion, and CI coverage improving from 8% to 50%; participants reported higher transparency and trust.

Causal Inference Pipeline for Employment Prediction(Korea Employment Information Service) | DoWhy · TabPFN

- Developed a custom estimator extending DoWhy's Estimator class (TabPFNestimator), integrating TabPFN—a transformer-based tabular foundation model—into DoWhy's standard pipeline to enable nonlinear causal effect estimation.
- Defined the shared causal analysis architecture for a multi-team project, coordinating data preprocessing, causal graph specification, and treatment effect estimation; implemented the common inference module used across teams to estimate causal effects of job-seeker characteristics on employment outcomes.

LLM Fine-tuning & Multi-GPU Training Seminar | [Github](#) | PyTorch · HuggingFace · DeepSpeed · Accelerate

- Designed and delivered a technical seminar covering end-to-end LLM supervised fine-tuning, from tokenization and label masking to GPU memory optimization, multi-GPU parallelism, and parameter-efficient training.
- Built a modular code repository with runnable examples for each training paradigm, including raw PyTorch SFT loops, SFTTrainer workflows, NCCL collective operation demos, DeepSpeed pipeline parallelism and switchable Accelerate YAML configs enabling comparison across distributed strategies without modifying training code.

HONORS & AWARDS

- K-Data Science Hackathon – 3rd Prize**

Nov. 2024

 - Designed a system integrating ML, NLP, and LLM to predict legislative bill approvals and provide automated feedback reports.
 - Achieved 91% training accuracy using LightGBM, addressing data imbalance with SMOTE and feature engineering for sponsor and societal context variables.
- KUBS CDTB Data Analytics for Business – 3rd Prize**

Nov. 2024

 - Designed a data-driven solution for optimizing reusable cup deposit collection system in Seoul.
 - Constructed and refined datasets engineered key variables, and modeled profit metrics.
 - Applied Maximum Covering Location Problem (MCLP) algorithms to optimize collection point locations and visualized profitability based on recovery rates and operating costs.
- KB Financial 6th Future Finance A.I. Challenge – 3rd Prize**

Aug. 2024

 - Designed a supply chain visualization service using LLMs to extract and analyze supplier, customer, and affiliate relationships from corporate disclosures, enabling intuitive stakeholder mapping and risk assessment.
 - Developed a sensitivity tracking algorithm to predict the impact of news events on related company stock performance, through real-time sentiment analysis.
- LangChain KR x NAVER Cloud Generative AI Service Development Competition - 3rd Prize**

June. 2024

 - Developed a chatbot returning likability scores and natural conversations, utilizing models as HyperClova X and KcELECTRA.

ACTIVITIES

- KUBIG, Korea University Data Science & Artificial Intelligence Student Union**

Jan. 2024 – Present

President & Deep Learning Team

Seoul, Korea

 - Led a 45+ member student organization; organized and managed semester-long deep learning study tracks, project teams, and internal competitions.
 - Participated as a team member in multiple data science and AI competitions through the union, earning four prizes across ML, NLP, LLM, and finance AI tracks (see Honors & Awards)

SKILLS & QUALIFICATIONS

Technical Skills	Python(PyTorch, HuggingFace, LangGraph, DeepSpeed, Accelerate, DoWhy), SQL, R, SAS
Language Proficiency	Native in Korean, Fluent in English(TOEFL 109), Intermediate in Japanese(JLPT N2)
Certifications	SQL Developer (SQLD), Advanced Data Analytics Semi-Professional (ADSP)